ECVOL-bench: An Open, Identity-Controlled Benchmark for Multimodal Earnings-Call Volatility Prediction

Anonymous Author(s)

Abstract

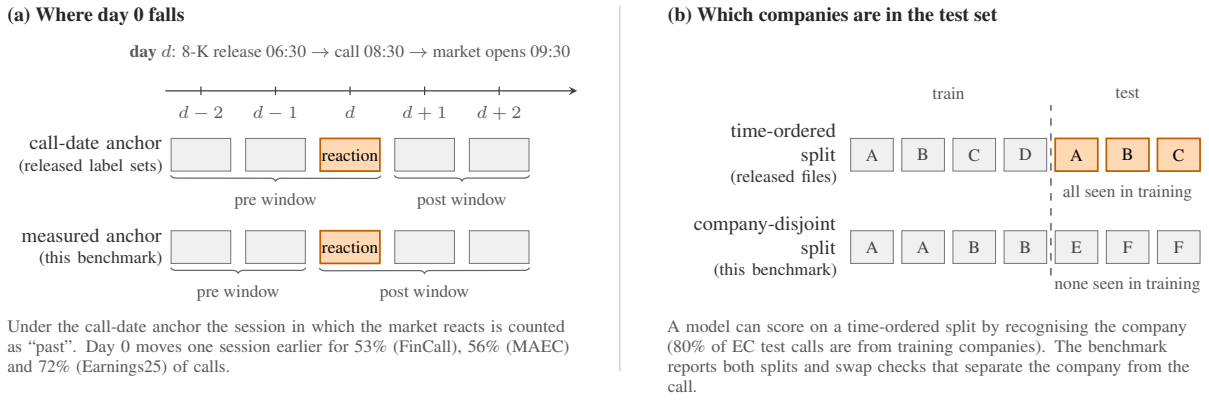
For about a decade, researchers have tried to predict how much a company's stock will move in the weeks after its quarterly earnings call, using what executives say on the call and how they sound. Many papers report that such models beat classical forecasts built from past price swings. We show that the test files these results share leave that claim untested. The files leave no gap between the training and test periods even though every target looks 30 days ahead. Between 44% and 80% of their test calls come from companies the model also saw in training. And they place "day zero" on the day of the call, although most companies release their numbers before the market opens that morning. We built ECVOL-bench to fix this. It adds company names and dates to a public audio corpus that ships without them, times each earnings release from SEC filings, and provides targets under both horizon definitions in use. It also gives splits that keep training and test apart in time and in company, and a set of identity checks that run on any file of predictions. With these checks in place, modern open text and audio models add nothing to a forecast built from past volatility alone. Text embeddings mostly identify which company is talking (89.5% probe accuracy), audio features do no better than a constant, and combining the two beats the classical forecast in two of eight cells, one of them a regime outlier. When we rerun five published models on their own data, the text models lose 12–82% in accuracy once test companies are unseen, while price-only models lose nothing. Scoring the frozen pipeline on 475 calls from late 2025, after every model's training cutoff, changes none of this. Content signal may still exist in these calls, but it has to be shown to survive these checks. The released benchmark, baselines and audit command make that the natural starting point for the next study.

1 Introduction

An earnings call is the one regular occasion on which a company's managers speak at length about their business. Since Qin and Yang (2019), a line of papers has used both the transcript and the audio of these calls to predict how volatile the stock will be over the following 3 to 30 trading days. HTML (Yang et al., 2020), MAEC (Li et al., 2020), VolTAGE (Sawhney et al., 2020a), DialogueGAT (Sang and Bao, 2022), KeFVP (Niu et al., 2023) and ECC Analyzer (Cao et al., 2024) all report beating classical forecasts. Two findings suggest that this evidence deserves a closer look. Yu et al. (2025) showed that transcript representations mostly tell the model *which company* is speaking, and that a simple forecast from the company's own past volatility beats every transcript model. Across NLP, a systematic review found that only 14% of reported scores were reproduced exactly (Belz et al., 2021), and in financial NLP discontinued data APIs and dead repository links make matters worse (Tatarinov et al., 2025).

We think both problems trace back to the same place. Nearly every paper in this line is scored on one set of label and split files, handed down from paper to paper. We audited those files (§2). They leave no gap between training and test dates even though each target runs 30 days past its call. Most test calls come from companies that are also in training, and no paper reports what happens when they are not. They define horizons in calendar days, which later papers describe as trading days. They put day 0 on the date of the call, although most companies publish their results before the market opens that day. The only open corpus with raw audio, FinCall-Surprise (Shu et al., 2026a), has no ticker, date or speaker fields at all, so the checks that would expose the company-identity problem cannot be run on it. Its successor, MM-FinEval (Shu et al., 2026b), adds tickers and dates but none of the checks. Figure 1 shows the two properties that matter most.

What we contribute. (1) **ECVOL-bench**. We recover company identities and dates for FinCall-Surprise (92.9% of calls, and a 60-call manual audit passes 60/60) and measure release times from 8-K filings. The benchmark ships three target families under both horizon definitions, splits with a 30-session gap and company-disjoint splits, checksums for every file, and a harness in which every table comes from one command (§2). (2) **An audit** of the shared label and split files (Table 1). (3) **Identity checks** (ticker probes, same-company and random content swaps, company-disjoint evaluation), released as a command that works on *any* prediction file. (4) **Results** for modern open-weight text and audio models against strong baselines (§4). (5) **A reproduction study** of the five runnable models in this line, on their own data, with the same checks, plus a code-availability audit of seventeen papers. (6) **A post-cutoff test** of the frozen pipeline on 475 calls from late 2025. Stages,



Under the call-date anchor the session in which the market reacts is counted as "past". Day 0 moves one session earlier for 53% (FinCall), 56% (MAEC) and 72% (Earnings25) of calls.

A model can score on a time-ordered split by recognising the company (80% of EC test calls are from training companies). The benchmark reports both splits and swap checks that separate the company from the call.

Figure 1: Two properties of the shared test files. (a) Most companies release results before the open, so anchoring day 0 on the call date leaves the session in which the market reacts inside the pre window. The benchmark anchors on the measured release time. (b) A time-ordered split keeps the same companies on both sides of the boundary. The benchmark adds a company-disjoint split and swap checks that make the difference measurable.

checks and confirmatory tests were recorded in a dated decision log, released with the benchmark, before the corresponding results existed.

Related work. The design responds to known pitfalls. Company identity leaks into transcript models (Yu et al., 2025), and language models recover it even when names are removed (Lopez-Lira et al., 2025), so we add identity checks. Language models also carry lookahead bias. Diagnostics and cutoff-bounded models now exist (Gao et al., 2025; Yan et al., 2026; Benhenda, 2026), but no earnings-call pipeline had been scored on calls after its cutoff, so we do that. Persistence, GARCH and HAR-RV (Corsi, 2009) are hard to beat and often missing from comparisons (Gu et al., 2020), so they anchor every table. Audio features encode speaker gender (Wang et al., 2024; Lin et al., 2024), so we report that confound. Yu et al. (2026) time earnings releases from 8-K filings to predict returns, but nobody had audited the volatility label sets. The strongest recent positive results (Chen et al., 2025; Cao et al., 2024) have not yet been evaluated with identity checks, and their code is not released. Whether they hold under these controls is an open question that the released audit command can settle.

2 The Shared Test Files and the Benchmark

What the field evaluates on. The EarningsCall (EC) benchmark of Qin and Yang (2019) and the two MAEC cohorts are scored on split and label files that VolTAGE and KeFVP redistribute. We audited those files (Table 1). The VolTAGE and KeFVP test files are byte for byte the same file, so the whole line of work shares one label set that was never recomputed from prices. Five of the six boundaries between training, validation and test have a zero-day gap (the sixth has three days) while the targets reach 30 days past each call. On the time-ordered splits the field reports, 80.4% (EC), 44.2% (MAEC-15) and 55.7% (MAEC-16) of test calls are from companies that also appear in training. Any time-ordered split allows that. The problem is that no paper reports the company-disjoint case, so the confound of Yu et al. (2025) has never been measured on these benchmarks. The original paper defines its horizons in calendar days, but most later work describes them as trading days (21 sessions rather than 30 at τ=30). Finally, the shipped MAEC labels correlate only 0.52–0.87 with calendar-day targets we recompute from prices under the same call-date anchor (0.25–0.66 at τ=3). We could not reproduce them under any single stated rule.

Benchmark / split	calls	overlap	embargo	label defect
EC (Qin & Yang)	560	80.4%	0 / 0 d	137 zeros
MAEC-15 (KeFVP)	765	44.2%	0 / 0 d	binary cols
MAEC-16 (KeFVP)	1400	55.7%	0 / 3 d	binary cols
ours: FinCall temporal	2200	91.3%	47 / 47 d	n/a
ours: FinCall ticker-disjoint	2433	**0.0%**	n/a	n/a
ours: MAEC temporal	2264	79.4%	101 / 48 d	n/a
ours: MAEC ticker-disjoint	2688	**0.0%**	n/a	n/a

Table 1: Audit of the shared benchmark files (Result Table 5R, regenerated by `ecvol audit substrate` from checksummed copies of the VolTAGE and KeFVP files, which are identical). Overlap is the share of test calls whose company is in training, and embargo is the gap train→val / val→test. Every time-ordered split allows overlap, ours included, but no paper in this line reports the disjoint case. Shipped MAEC labels correlate 0.52–0.87 with our recomputed calendar-day, call-date-anchored targets at $\tau \geq 7$.

Corpora and company identity. **FinCall-Surprise** (Shu et al., 2026a) is the primary corpus, with 2,688 calls from 2019–2021 and MP3 audio, transcripts and slides. It ships four fields per record and no ticker, date, CIK or speaker names. We recover the company and the date from the transcript text and the slide metadata,

check them against price data, and release the table with its 80 manual corrections. 2,496 calls resolve (92.9% overall, 95.3% of the earnings-type calls), and a 60-call manual audit passes 60/60. **MAEC** (Li et al., 2020) is the secondary corpus, with about 3,443 calls from 2015–2018, sentence-aligned audio features and no raw audio. **Earnings25** (Jiang et al., 2026) has 514 S&P 500 calls from July–December 2025, with webcast MP3 at 64 kbps for 281 calls and at most 24 kbps for 232. It postdates every model's training data, and 475 of its calls map to a ticker and join to prices.

Targets. Daily adjusted prices come from a free source and were checked against Tiingo on 50 tickers. Let r_t be the return on session t, counted from day 0, the last session before the call's information is public. The main target is $v_{\text{post}}(\tau) = \ln \sqrt{\frac{1}{n}\sum_{t=1}^{n}(r_t - \bar{r})^2}$ for $\tau \in \{3, 7, 15, 30\}$, and $v_{\text{pre}}(\tau)$ is the same quantity over the sessions ending at day 0. Under the **trading-day** definition $n = \tau$. Under the **calendar-day** definition of Qin and Yang (2019) the window holds every session within τ calendar days, so $n \approx 0.7\tau$, and at τ=3 a Thursday or Friday call has at most one session. We drop any window with fewer than two sessions and record the reason. That removes 42.5% of FinCall's calls at τ=3 calendar days, for which the released label sets still carry a value. The two definitions are not interchangeable. v_{post} correlates only 0.78 between them at τ=3, and on the temporal splits the persistence MSE is 44–63% higher there under calendar days. A published MSE therefore means little unless its definition is stated (Appendix Table 9). Every result below uses trading days, and the definition is a column in every result table. We add two targets that are harder to game with company identity. The **change** $\Delta v = v_{\text{post}} - v_{\text{pre}}$ subtracts the company's own level, and the **HAR-residual** subtracts a HAR-RV forecast. After joining to prices, FinCall has 2,433 calls with a target (97.5%) and MAEC 2,688 (78.6%, the rest lacking usable price history).

When the information became public. Almost no transcript states a clock time, so we take the moment the information became public from EDGAR. It is the acceptance time of the earliest 8-K filing with Item 2.02 (the earnings release) within three days before to one day after the call. That covers 89.3% of FinCall, 68.6% of MAEC and 99.8% of Earnings25. The rest fall back, in recorded order, to Earnings25's call-start time, the before/after-market flag of Yu et al. (2025), and an after-hours assumption. A release before the open makes the call date itself day 1. Compared with the call-date anchor that every released label set uses, **day 0 moves for 53% of FinCall, 56% of MAEC and 72% of Earnings25 calls**, almost always one session earlier. Under the published anchor, the session in which the market reacts therefore sits in the *pre* window for most calls. On the measured anchor the persistence MSE is up to 11% higher on FinCall, 25% on MAEC and 27% on Earnings25, and Δv gains a positive mean (on FinCall $+0.52$ at τ=3 and $+0.10$ at τ=30). The call-date targets ship as a labelled variant.

Splits and release. *Temporal* splits (about 70/10/20 in date order) leave a 30-trading-day gap between segments so that no target window crosses a boundary. *ticker-disjoint* splits keep every validation and test company out of training. Their overlap is 0.0%, against 91.3% (FinCall) and 79.4% (MAEC) on our own temporal splits. The harness fails any feature that reads price data after the call's as-of time, and leakage assertions run in CI. The release contains all of the above, the derived-feature archives, the harness, and every result CSV with its run record (resolved config, git commit, seeds and file hash). Raw audio and transcripts are never redistributed.

3 Models, Protocol and Checks

Models, from simplest to richest. We organise our own models in stages. **Stage 0** holds the classical forecasts, namely persistence ($\hat{v} = v_{\text{pre}}$), the training-set mean of the target (a constant), EWMA (λ=0.94), HAR-RV (Corsi, 2009) and GARCH(1,1). **Stage 1** is a LightGBM model on past-volatility features, transcript length, number of Q&A turns and a **company fixed effect**, our identity-only baseline. **Stage 2** uses the transcript through BGE-M3 (BAAI, 2024) embeddings pooled per section (1024-d), FinBERT (ProsusAI, 2019) sentiment aggregates and surface statistics. **Stage 3** uses the audio, resampled to 16 kHz mono. Its features are eGeMAPSv02 functionals (Eyben et al., 2010) and WavLM-Large (Chen et al., 2021) and emotion2vec+ (Ma et al., 2023) embeddings over 30-second windows, averaged per call. **Stage 4** combines modalities by gated fusion or by stacking (a ridge over the text, audio and Stage-1 predictions, fit on the validation set). From Stage 2 on, every model is reported with and without a block of past-volatility features (v_{pre} at all horizons plus the HAR-RV inputs). The prediction heads are a ridge regression, with its penalty chosen on validation, and a small MLP run with five seeds. The MLPs are unstable across seeds (standard deviation of R^2_{OOS} up to 6.8), so the main grid uses one deterministic model per stage, chosen by a rule recorded in the decision log. MLP cells are in Appendix A.

Metrics and tests. We report MSE on log realised volatility and, as the headline metric, R^2_{OOS} relative to persistence (Gu et al., 2020), defined as $1 - \sum(y - \hat{y})^2 / \sum(y - \hat{y}_{\text{persist}})^2$. Because Δv has a positive mean on the measured anchor, persistence (which predicts Δv=0) is a weak reference there, so every Δv claim is also compared with the constant model. Diebold–Mariano tests (Diebold and Mariano, 1995) with the

small-sample correction compare per-call squared errors against persistence and against the previous stage. The confirmatory comparisons (each stage against Stage 1 on Δv across horizons, Stage 4 against the best single modality, and the published split against the controlled split for each reproduced model) are Holm-corrected. All other analyses are exploratory.

Identity checks. We use four. A *company-only* model predicts each company's training mean and shows how far identity alone gets. Two *swaps* replace each test call's transcript or audio, either with another call by the same company or with a random call from the corpus. If the same-company swap leaves a model's score unchanged, the model was reading identity, and if the random swap also leaves it unchanged, the model was reading nothing. A linear *probe* measures how much identity the frozen call embedding carries by trying to recover the ticker from it. Finally, every table shows ticker-disjoint results next to temporal ones. The command `ecvol audit predictions` runs the prediction-side versions of the company-only and swap checks on any file with one row per (call, ticker, date, split, horizon, truth, prediction). It also reports split integrity, seen versus unseen companies, the between-company share of prediction variance and company-clustered significance. A published result can therefore be checked from the one file its authors already have.

Reproductions and post-cutoff tests. Of seventeen papers in this line, one releases code that runs as released (Same-Company-Same-Signal), seven release code that needs repairs or inputs that were never shipped, and nine release nothing usable (Table 15). We rerun the five with usable code (HTML, KeFVP, Same-Company-Same-Signal, Sawhney et al., 2020b, DialogueGAT) *on their own data*, using the authors' code wherever it can be made to start and a line-by-line port where it cannot. Every patch is in a released ledger. We first compare with the published number and then rerun the same model under a gapped split and a ticker-disjoint split of the same data. Earnings25 allows two more tests, namely the frozen Stage-3 audio models rerun by bitrate, and the whole FinCall-trained pipeline scored, with no retraining, on calls that postdate every model's training cutoff.

4 Results

The tables in this section use FinCall-Surprise, the Δv target and the trading-day definition. MAEC, the level target, MSE, the calendar-day definition and every other model are in Appendix A and in the supplement, regenerated from the released CSVs.

The baseline to beat. HAR-RV is strong everywhere except one cell (Table 2). On the FinCall temporal split at τ=30, the training period includes the COVID spike and the test year (2021) is calm. There, every long-horizon forecast of the volatility level does worse than persistence, ours and every content model alike. The same models do fine on the ticker-disjoint split and on MAEC, so we treat this as a regime effect and note it. On the measured anchor a constant (the training mean) already scores 0.17, 0.16, 0.06 and 0.09 on the temporal split and 0.20, 0.20, 0.13 and 0.04 on the ticker-disjoint split. That is the reference for every Δv claim below.

Model	τ=3	τ=7	τ=15	τ=30
Temporal split (n=483)				
S0 train mean	0.165	0.158	0.060	0.088
S0 HAR-RV	0.444	0.372	0.138	−0.285
S0′ past-vol ridge	0.412	0.314	0.094	0.113
S1 GBDT (ticker FE)	0.340	0.249	−0.126	−0.948
S2 text ridge + vol	0.333	0.285	0.080*	−0.077*
S3 audio ridge + vol	0.353	0.268	0.033*	0.050*
S4 stack fusion	0.406*	0.347*	0.066*	0.100*
ticker-disjoint split (n=484)				
S0 train mean	0.200	0.201	0.132	0.035
S0 HAR-RV	0.570	0.491	0.349	0.345
S0′ past-vol ridge	0.552	0.469	0.345	0.318
S1 GBDT (ticker FE)	0.494	0.440	0.292	0.237
S2 text ridge + vol	0.399†	0.375	0.256	0.301
S3 audio ridge + vol	0.442	0.364	0.241	0.217
S4 stack fusion	0.531	0.465	0.340	0.396*

Table 2: The ladder on FinCall-Surprise. R^2_{OOS} vs. persistence on Δv (test). Persistence is 0 by construction, and the train-mean intercept is the featureless floor. One pre-registered canonical model per stage (the Stage-3 ridge uses WavLM and eGeMAPS). An asterisk or dagger marks a Holm-corrected Diebold–Mariano result better or worse than Stage 1 ($p < 0.05$). Adding text or audio to the past-volatility ridge (S0′) lowers R^2_{OOS} in every cell. The full grid, MAEC, level-v, MSE and the MLP heads are in Appendix A.

Text models read the company, not the call (Table 3, Figure 2). The text-only ridge stays within 0.14 of the constant on both splits and is below HAR-RV by at least 0.09 in seven of eight cells (the eighth is the COVID cell). The MLPs overfit. Adding text features to the past-volatility ridge makes it *worse* in all eight cells. A linear probe recovers the ticker from the frozen call embedding 89.5% of the time on FinCall and 53.9% on MAEC. On the temporal split, where test companies also appear in training, giving each test call the transcript of *another call by the same company* changes the score by less than 0.03, while a random transcript hurts. The model is reading which company is speaking. On the ticker-disjoint split the same-company swap does hurt at $\tau \geq 15$, so there is some call-specific signal for companies the model has never seen. That signal stays below the past-volatility baseline, and it disappears once past volatility is in the model (0.301 against 0.318 at τ=30). MAEC shows the same pattern.

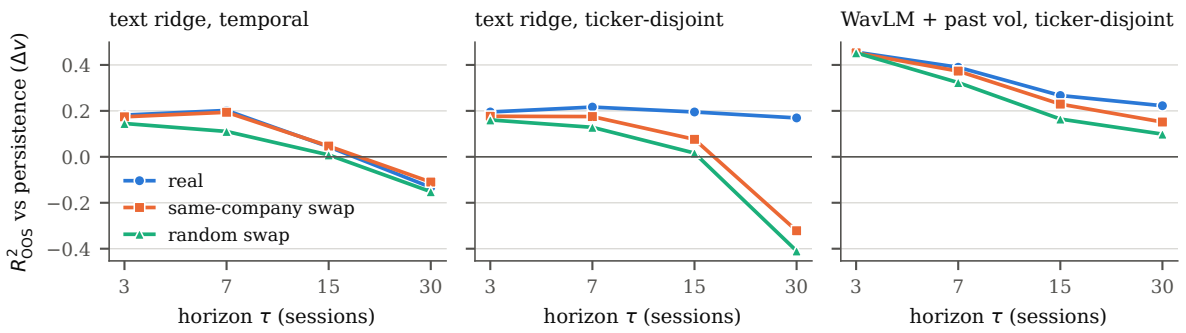


Figure 2: Content swaps. R^2_{OOS} vs. persistence on Δv (test) when each test call keeps its own transcript or audio (real), receives another call by the same company, or receives a random call. Where companies recur in training (left), the same-company swap leaves the text model unchanged. On the ticker-disjoint split (middle) the model loses something at long horizons, but stays below the past-volatility ridge of Table 2. The audio model with past volatility (right) keeps its score under a random swap at τ=3. Data from Table 3.

Identity probe (linear, frozen call embedding)	accuracy	×chance
BGE-M3 text, FinCall (357 tickers)	89.5%	319×
BGE-M3 text, MAEC (902 tickers)	53.9%	487×
WavLM audio, FinCall (356 tickers)	76.3%	271×
emotion2vec+ audio, FinCall (356 tickers)	29.4%	105×

Content swap (R^2_{OOS} on Δv)	τ=3	τ=7	τ=15	τ=30
Text ridge, temporal split				
real	0.182	0.202	0.044	−0.133
same-company swap	0.174	0.194	0.047	−0.110
random swap	0.145	0.110	0.008	−0.152
Text ridge, ticker-disjoint split				
real	0.195	0.217	0.195	0.169
same-company swap	0.176	0.175	0.076	−0.322
random swap	0.161	0.128	0.016	−0.410
WavLM ridge + past-vol, ticker-disjoint split				
real	0.456	0.390	0.267	0.222
same-company swap	0.453	0.373	0.229	0.152
random swap	0.453	0.323	0.164	0.099

Table 3: Identity checks. The top block shows that frozen embeddings identify the company far above chance. The bottom block shows that where companies recur (temporal), giving a test call another call by the *same* company barely changes the text model, while a random call hurts. On the ticker-disjoint split the text model's remaining signal at $\tau \geq 15$ sits below the past-volatility baseline of Table 2.

Audio adds nothing to past volatility. Audio-only ridges on eGeMAPS, WavLM-Large and emotion2vec+ stay within 0.10 of the constant at every horizon on both splits and never beat it by more than 0.02. Adding WavLM to the past-volatility ridge makes it worse in all eight cells. In the audio-swap check (Table 3, bottom), the model with past volatility keeps its score at τ=3 when every test call is given a *random other company's* audio (0.456 to 0.453). At longer horizons the swap costs more. Even so, the model with real audio stays below the ridge with past volatility alone, so whatever the audio contributes does not add to the baseline. A probe still recovers the ticker from pooled WavLM embeddings 76.3% of the time. A gender check (dominant-speaker pitch proxy) finds no error gap (test MSE 0.243 against 0.239). Rerun on Earnings25 by bitrate, the gain from real over randomly swapped audio never exceeds 0.03, is negative in 7 of 12 cells, and is no larger at 64 kbps than at lower bitrates (Appendix Table 13). The FinCall result is not an artefact of its 40 kbps audio.

Combining modalities. Gated fusion is below the ridge heads with past volatility in 15 of 16 cells. The stacking model is the best content model but stays below HAR-RV in six of eight cells. It beats HAR-RV only in the COVID cell, where it scores 0.100 and is still below the past-volatility ridge at 0.113, and at ticker-disjoint τ=30, where it scores 0.396 against 0.345 (Holm p=0.003 against Stage 1, with no pre-registered test against HAR-RV). It beats the text and audio stages in seven of eight cells and the past-volatility ridge in two. So a multimodal model is not better than the best single-input model when that input is past volatility. The condition we set in advance for a positive paper, gains surviving the checks on at least two of four horizons, was not met.

Published models under the checks (Table 4, Figure 3). Among the five papers we could run, the results that reproduce belong to price-only or same-company models (Appendix A.3). Same-Company-Same-Signal reproduces from its shipped data. TSMixer matches to 10^{-5} and TMLP within 0.03, including their own within-company swap result. Their training-free same-company history comes within 0.05 of KeFVP's published MSE at every horizon on EC and MAEC-15, with no text and no training. That model is undefined once companies are held out. KeFVP is a price-series model conditioned on text, with its audio branch disabled in its own code. It reproduces seven of eight MAEC cells within one published standard deviation. Its EC row, run without the knowledge-enhanced embeddings that are no longer available, is above all four published cells and we do not count it. HTML's released driver shuffles features and labels independently

Model (data), change in test MSE	τ=3	τ=7	τ=15	τ=30
Companies held out (ticker-disjoint vs. the model's own split)				
HTML text (EC)	+18%	+55%	+55%	+82%
HTML text+audio (EC)	+20%	+44%	+61%	+70%
Sawhney ensemble (EC)	+13%	+28%	+41%	+38%
SCSS TMLP, text (DEC)	+12%	+23%	+25%	+27%
SCSS TSMixer, price (DEC)	+7%	+5%	−6%	−8%
KeFVP, text+price (MAEC-15)	−3%	+4%	+3%	−1%
KeFVP, text+price (MAEC-16)	−9%	−3%	−10%	−4%
30-session embargo vs. the model's own split				
HTML text (EC)	+10%	+3%	−3%	+11%
Sawhney ensemble (EC)	+5%	−6%	−2%	−9%
SCSS TMLP (DEC)	−4%	+1%	0%	+3%
HTML text, transcript replaced by (own split, 97 calls)				
another call, *same* company	+1%	+4%	+3%	+2%
a call of a *different* company	+9%	+11%	+21%	+23%

Table 4: The field's runnable models under our controls (Result Table 6R-C). Each row keeps the configuration of Appendix Table 11 and changes only the split. Holding companies out costs every text model 12–82% and leaves the price-dominated models unchanged, with held-out training sets as large (EC) or cut by the same third (DEC, MAEC). Absolute values are in Appendix Table 12.

without a seed, so the released code does not reproduce the published table. With the authors' model classes unchanged, the published recipe reaches the published 15- and 30-day values on no split. The published ensemble of Sawhney et al. (2020b) is matched at 7 days and beyond. In the released scripts, however, the text and audio branches are untrained after the prescribed epochs and the mixing weights are chosen on the test set, so the match comes from the finance branch. Under our checks (Table 4), holding companies out costs the three text models 12–82% in MSE, while TSMixer and KeFVP lose nothing. Training-set size does not explain the difference, since the held-out training sets are as large (EC) or cut by the same third (DEC, MAEC) for both groups. The gap between training and test moves models by at most 13%. HTML's own trained model changes by at most 4% when a test transcript is swapped for another call by the same company, against 9–23% for a different company's call. On this evidence, the HTML text model is reading which company is speaking.

After the cutoff (Table 5). We scored the FinCall-trained models, unchanged, on 475 Earnings25 calls from late 2025, after every model's training cutoff and on companies none had seen. On the level target, no model that was competitive before the cutoff got worse. HAR-RV rose from 0.44, 0.37, 0.14 and −0.29 to 0.60, 0.54, 0.35 and 0.08, and the content ridges rose with it (only the fusion ridge is unstable, at τ=30). Levels are not comparable across the cutoff, since the years and companies differ, but the ordering is, and it holds. HAR-RV beats every content model at every horizon, and no content model that was competitive before the cutoff collapses after it. Under the call-date anchor of the published label sets, the same frozen models collapse

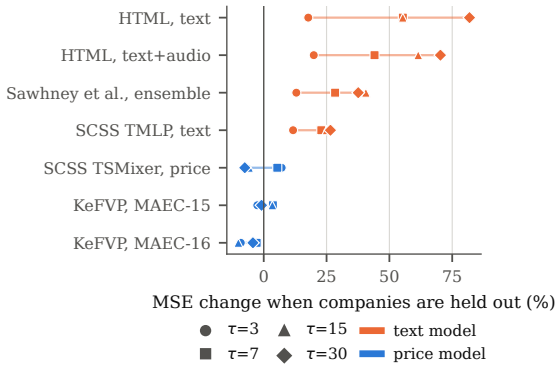


Figure 3: Published models with companies held out. Change in test MSE when the same model is scored on a ticker-disjoint split of its own data, one marker per horizon (data from Table 4). The three text models lose 12–82%, and the two price-based models lose nothing.

Model	cutoff	τ=3	τ=7	τ=15	τ=30
S0 train mean	before	0.36	0.14	−0.37	−1.05
	after	0.58	0.47	0.15	−0.29
S0 HAR-RV	before	0.44	0.37	0.14	−0.29
	after	0.60	0.54	0.35	0.08
S2 text ridge + vol	before	0.43	0.34	0.03	−0.45
	after	0.57	0.50	0.25	−0.20
S3 WavLM ridge + vol	before	0.38	0.24	−0.09	−0.66
	after	0.57	0.43	0.15	−0.32
S4 fusion ridge + vol	before	0.41	0.32	−0.02	−0.63
	after	0.50	0.46	0.20	−1.93

Table 5: Frozen-pipeline post-cutoff evaluation (Result Table 7). Heads fit once on FinCall's training split and scored unchanged before the cutoff (the 2021 FinCall test year) and after it (475 Earnings25 calls, late 2025), on the level target, R^2_{OOS} vs. persistence. The ordering survives the cutoff. Bootstrap confidence intervals and the Δv rows are in the released CSV.

before the cutoff. Text plus past volatility on Δv falls to between −1.7 and −3.6 at 3 to 15 days (Appendix Table 14), because the reaction session then sits inside the past-volatility features.

5 Discussion

Four things hold across the whole set of models. Classical volatility forecasts are hard to beat, and measuring against persistence exposes a regime failure that raw MSE hides. Text features add nothing to past volatility, because the embeddings identify the company. Where companies recur in training, swapping in another call by the same company changes nothing. Audio features add nothing either, at 40 kbps or 64 kbps, even though they too identify the company, and combining the two does not help. None of this could be measured with the files the field has been using, which is the gap the benchmark fills. The same checks on five published models give the same answer. What reproduces is price-only models or same-company history, and the text models lose their edge as soon as the company is

new. The strongest recent positive results (Chen et al., 2025; Cao et al., 2024) could not be run through these checks because their code is not released. The `audit predictions` command makes that check cheap for anyone with a file of predictions. We do not claim that transcripts and audio carry no information about a company's risk. We claim that the benchmarks in use cannot tell that information apart from knowing which company is speaking. Once they can, nothing yet built beats a forecast from past volatility. That is the baseline the next model has to clear.

6 Recommendations and Outlook

The results above are not an argument against using earnings calls. They are an argument about what an evaluation has to control for before a content effect can be claimed. We suggest that future work on this task report six things. A **company-disjoint** split next to the temporal one, since identity carries most of the apparent signal. A **gap** between training and test dates of at least the longest horizon. **Past-volatility baselines** (persistence and HAR-RV, plus a constant on change targets) as the reference for every claim. The **same-company and random swaps**, or the released audit command, so that a gain can be attributed to the call rather than to the company. **Day 0 measured** from the release time rather than the call date. And the **horizon definition**, calendar or trading days, stated with every number. ECVOL-bench provides each of these for FinCall-Surprise and MAEC, and the audit command applies the checks to any prediction file. The release also ships the processed inputs as one archive per corpus, with checksums. Each archive holds the targets under both definitions and anchors, the splits, the pooled text and audio features, and the identity and timing tables. A quickstart script goes from an archive to the past-volatility baseline, a prediction file and the audit report in a few seconds, so a new model can be compared against the same floor and checks on the first day.

There is also reason to expect that content signal can be found once it is looked for in the right place. Measured release times make **event-window targets** possible, such as the volatility of the first sessions after the release, where language about guidance and surprise should matter most and where same-company history explains least. **Structured features** extracted from the call, such as guidance changes and the gap between the quantitative surprise and management's framing of it, can be audited one field at a time, unlike pooled embeddings. They can also be combined with the fast numeric signal that Yu et al. (2026) show arrives first. **Audio** has so far been tested only on lossy webcast transcodes without speaker diarisation, so recordings at source quality with resolved speakers are the fair test of prosody. **Larger company-disjoint corpora** are now within reach, since MM-FinEval (Shu et al., 2026b) ships ticker and date fields and the checks here transfer to it directly. Each of these can be evaluated against the released baselines on the first day, which is what the benchmark is for.

Limitations

One post-cutoff quarter. The post-cutoff test rests on 475 calls from a single calm quarter and a different universe from the training corpus, so it bounds contamination in the ordering of models, not in their absolute scores. **Reproductions run on their own data**, not on ECVOL-bench, to keep "the model fails" separable from "the model does not transfer". Two carry known costs. KeFVP's released EC embeddings are unavailable upstream, and DialogueGAT's corpus was never released, so our port stops at its protocol. **MLP heads are seed-unstable** (standard deviation of R^2_{OOS} across seeds up to 6.8 on the level target and 1.1 on Δv), which is why the grid uses deterministic models. MLP cells in the appendix are seed means with that caveat. **Short-horizon R^2_{OOS} inflation.** The persistence denominator is noisiest at τ=3 (its MSE is about 2.8 times that at τ=7), so short-horizon gains are easier to show. **Timestamps.** 10.7% of FinCall and 31.4% of MAEC calls (13.0% of the MAEC calls that carry targets) have no Item 2.02 filing in the window and use a recorded fallback tier, and intraday releases are treated as after-hours. Results under the call-date anchor of prior label sets are not interchangeable with ours, and the call-date targets are shipped as a variant. **Regime sensitivity.** FinCall-Surprise spans COVID. The temporal τ=30 failure shows that long-horizon level forecasts are regime-sensitive, which is why we emphasise Δv and per-year breakdowns. **MAEC coverage.** Its 78.6% price-join rate (2015–2018 delistings) makes it a secondary corpus, and it ships no raw audio. **Source audio.** Every FinCall file is a 40 kbps MP3 transcode and Earnings25 is 64 kbps or lower, so no result here is established on lossless audio. **No diarisation.** Audio is pooled per call, and the gender analysis uses a dominant-speaker pitch proxy. **Identity reconstruction is imperfect.** 7.1% of FinCall calls remain unresolved and are excluded with a reason code.

Ethics Statement

Data and licensing. We use openly licensed corpora (FinCall-Surprise under Apache-2.0, MAEC under CC-BY-SA-4.0, Earnings25 under CC-BY-4.0) and open-weight models. Derived features inherit and state their upstream licence, and raw audio or transcripts from sources with unclear terms (EC, DEC, the reproduced repositories) are never redistributed. Calls are public corporate disclosures, and we release no speaker-level audio or identity, only pooled per-call features. **Gender confound.** Audio features encode speaker gender and female executives are underrepresented in these corpora

(Wang et al., 2024; Lin et al., 2024; Mathur et al., 2022), so we report a gender-confound diagnostic rather than silently propagate the bias. **Intended use.** This is a forecasting-research benchmark, not investment advice. Predictions are noisy and regime-sensitive and should not drive automated trading. **Reproducibility.** Given the documented reproducibility problems in financial NLP (Tatarinov et al., 2025), the environment is pinned, every data file is checksummed, every result CSV regenerates byte for byte from its command and run manifest, and the curated tables copy cells from those CSVs.

References

BAAI. BGE-M3 sentence embeddings. Hugging Face (MIT), 2024. URL https://huggingface.co/BAAI/bge-m3.

Anya Belz, Shubham Agarwal, Anastasia Shimorina, and Ehud Reiter. A systematic review of reproducibility research in natural language processing. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume (EACL)*, pages 381–393, 2021. URL https://aclanthology.org/2021.eacl-main.29/.

Mostapha Benhenda. Look-ahead-bench: a standardized benchmark of look-ahead bias in point-in-time LLMs for finance. arXiv:2601.13770, 2026. URL https://arxiv.org/abs/2601.13770.

Yupeng Cao et al. ECC analyzer: Extracting trading signal from earnings conference calls using large language models for stock volatility prediction. In *Proceedings of the 5th ACM International Conference on AI in Finance (ICAIF)*, 2024. URL https://arxiv.org/abs/2404.18470.

Sanyuan Chen et al. WavLM: Large-scale self-supervised pre-training for full stack speech processing. Hugging Face (MIT): microsoft/wavlm-large, 2021. URL https://huggingface.co/microsoft/wavlm-large.

Xiaoliang Chen et al. The sound of risk: A multimodal physics-informed acoustic model for forecasting market volatility and enhancing market interpretability. arXiv:2508.18653, 2025. URL https://arxiv.org/abs/2508.18653.

Fulvio Corsi. A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 2009. doi: 10.1093/jjfinec/nbp001.

Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 1995. doi: 10.1080/07350015.1995.10524599.

Florian Eyben, Martin Wöllmer, and Björn Schuller. openSMILE / eGeMAPS acoustic feature extraction. github.com/audeering/opensmile-python, 2010. URL https://github.com/audeering/opensmile-python.

Zhenyu Gao, Wenxi Jiang, and Yutong Yan. Detecting lookahead bias in LLM forecasts. arXiv:2512.23847, 2025. URL https://arxiv.org/abs/2512.23847.

Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *The Review of Financial Studies*, 2020. doi: 10.1093/rfs/hhaa009.

Denglin Jiang, Haoran Zhou, Anshul Wadhawan, Brendan Fahy, Vinay Ramesh, David Weisberg, Dmitriy Derkachevskiy, Helen Sheehan, Srivas Prasad, and Michele Franceschini. Earnings25: A comprehensive 500-hour speech benchmark for finance. Interspeech 2026; arXiv:2607.23813; Zenodo record 10.5281/zenodo.18762168 (CC-BY-4.0), 2026.

Jiazheng Li, Linyi Yang, Barry Smyth, and Ruihai Dong. MAEC: A multimodal aligned earnings conference call dataset for financial risk prediction. In *Proceedings of the 29th ACM International Conference on Information and Knowledge Management (CIKM)*, 2020. doi: 10.1145/3340531.3412879.

Yi-Cheng Lin et al. Emo-bias: A large scale evaluation of social bias on speech emotion recognition. arXiv:2406.05065, 2024. URL https://arxiv.org/abs/2406.05065.

Alejandro Lopez-Lira, Yuehua Tang, and Mingyin Zhu. The memorization problem: Can we trust LLMs' economic forecasts? arXiv:2504.14765, 2025. URL https://arxiv.org/abs/2504.14765.

Ziyang Ma et al. emotion2vec: Self-supervised pre-training for speech emotion representation. Hugging Face: emotion2vec/emotion2vec_plus_large, 2023. URL https://huggingface.co/emotion2vec/emotion2vec_plus_large.

Puneet Mathur et al. DocFin: Multimodal financial prediction and bias mitigation using semi-structured documents. In *Findings of the Association for Computational Linguistics: EMNLP 2022*, 2022. URL https://aclanthology.org/2022.findings-emnlp.139/.

Hao Niu et al. KeFVP: Knowledge-enhanced financial volatility prediction. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023. URL https://aclanthology.org/2023.findings-emnlp.770/.

ProsusAI. FinBERT: Financial sentiment analysis with bert. Hugging Face, 2019. URL https://huggingface.co/ProsusAI/finbert.

Yu Qin and Yi Yang. What you say and how you say it matters: Predicting stock volatility using verbal and vocal cues. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2019. URL https://aclanthology.org/P19-1038/.

Yunxin Sang and Yang Bao. DialogueGAT: A graph attention network for financial risk prediction by modeling the dialogues in earnings conference calls. In *Findings of the Association for Computational Linguistics: EMNLP 2022*, 2022. URL https://aclanthology.org/2022.findings-emnlp.117/.

Ramit Sawhney, Piyush Khanna, Arshiya Aggarwal, Taru Jain, Puneet Mathur, and Rajiv Ratn Shah. VolTAGE: Volatility forecasting via text audio fusion with graph convolution networks for earnings calls. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020a. URL https://aclanthology.org/2020.emnlp-main.643/.

Ramit Sawhney, Puneet Mathur, Ayush Mangal, Piyush Khanna, Rajiv Ratn Shah, and Roger Zimmermann. Multimodal multi-task financial risk forecasting. In *Proceedings of the 28th ACM International Conference on Multimedia (MM)*, 2020b. doi: 10.1145/3394171.3413752.

Dong Shu, Yanguang Liu, Huopu Zhang, and Mengnan Du. FinCall-Surprise: A large scale multi-modal benchmark for earning surprise prediction. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13357–13370, 2026a. URL https://aclanthology.org/2026.acl-long.610/. Data: github.com/Tizzzzy/FinCall-Surprise (Apache-2.0).

Dong Shu, Yanguang Liu, Huopu Zhang, Saisai Hu, Haiyan Zhao, Hekun Huang, and Mengnan Du. MM-FinEval: A multi-task multimodal benchmark for real-world financial forecasting. arXiv:2609.38523, 2026b. URL https://arxiv.org/abs/2609.38523.

Nikita Tatarinov et al. Language modeling for the future of finance: A survey into metrics, tasks, and data opportunities. arXiv:2504.07274, 2025. URL https://arxiv.org/abs/2504.07274.

Wang et al. AMA-LSTM: Pioneering robust and fair financial audio analysis for stock volatility prediction. NAACL 2024 Industry Track; arXiv:2407.18324, 2024. URL https://arxiv.org/abs/2407.18324.

Yutong Yan et al. DatedGPT: Preventing lookahead bias in large language models with time-aware pretraining. arXiv:2603.11838, 2026. URL https://arxiv.org/abs/2603.11838.

Linyi Yang, Tin Lok James Ng, Barry Smyth, and Ruihai Dong. HTML: Hierarchical transformer-based multi-task learning for volatility prediction. In *Proceedings of The Web Conference (WWW)*, 2020.

Ding Yu, Zhuo Liu, and Hangfeng He. Same company, same signal: The role of identity in earnings call transcripts. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 18403–18422, 2025. URL https://aclanthology.org/2025.findings-acl.946/.

Ding Yu, Zhuo Liu, Hao Zhang, and Hangfeng He. Fast numbers, slow language: Bridging quantitative and qualitative earnings signals. arXiv:2606.29734, 2026. URL https://arxiv.org/abs/2606.29734.

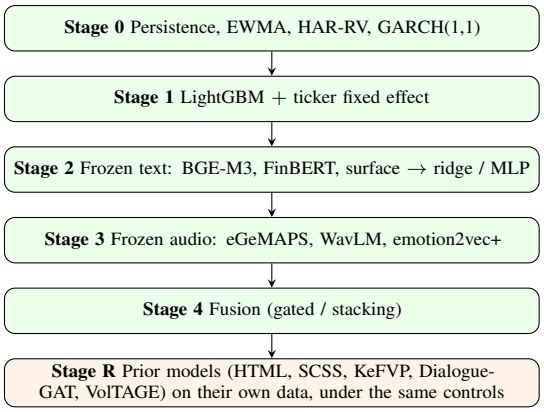


Figure 4: The modelling ladder. From Stage 2 up, every model also carries past-volatility covariates (and is reported with them ablated). Every stage runs through the identity-control suite. Green = complete, orange = pending.

A Extended Results

This appendix holds the full set of models behind each main-text table, the horizon-definition comparison, the reproduction study in detail, the Earnings25 bitrate study, the call-date-anchor variant of the post-cutoff test, and the code-availability audit. The complete generated grids (Result Tables 1–4, every corpus, target, metric, split and horizon) are in the separate supplement. Every number traces to a released result CSV with a run record, and the curated tables copy cells from those CSVs (`REPRODUCE.md` lists the command behind each).

A.1 All models of Stages 0–3

Tables 6, 7 and 8 give the full Stage-0/1, Stage-2 and Stage-3 model sets on the Δv target, including the past-volatility-only and MLP heads omitted from Table 2.

Model	τ=3	τ=7	τ=15	τ=30
Temporal split (n=483)				
Persistence	0.000	0.000	0.000	0.000
EWMA	0.465*	0.389*	0.259*	0.019
HAR-RV	0.444*	0.372*	0.138*	-0.285*
GARCH(1,1)	0.472*	0.463*	0.298*	0.023
GBDT (ticker FE)	0.340*	0.249*	-0.126	-0.948*
Ticker-disjoint split (n=484)				
Persistence	0.000	0.000	0.000	0.000
EWMA	0.520*	0.379*	0.142*	0.112*
HAR-RV	0.570*	0.491*	0.349*	0.345*
GARCH(1,1)	0.561*	0.484*	0.334*	0.298*
GBDT (ticker FE)	0.494*	0.440*	0.292*	0.237*

Table 6: Baselines. R^2_{OOS} vs. persistence on the Δv target, FinCall-Surprise (test). HAR-RV is strong except on the COVID-regime temporal τ=30 cell. Asterisks mark Diebold–Mariano significance vs. persistence ($p < 0.05$).

Model	τ=3	τ=7	τ=15	τ=30
Temporal split (n=483)				
Ridge (text)	0.182*	0.202	0.044*	-0.133*
Ridge (past-vol)	0.412*	0.314*	0.094*	0.113*
Ridge (text+vol)	0.333	0.285	0.080*	-0.077*
MLP (text)	0.013*	0.069*	-0.253	-1.202
Ticker-disjoint split (n=484)				
Ridge (text)	0.195*	0.217*	0.195*	0.169
Ridge (past-vol)	0.552*	0.469	0.345	0.318*
Ridge (text+vol)	0.399*	0.375	0.256	0.301
MLP (past-vol)	0.577*	0.508*	0.381*	0.356*

Table 7: Text content (Stage 2). R^2_{OOS} vs. persistence on the Δv target, FinCall-Surprise (test). The text-only head does not separate from the past-volatility head, and adding text to past volatility does not help. Asterisks mark Diebold–Mariano significance vs. the Stage-1 ticker-FE GBDT ($p < 0.05$). Full head set in Appendix A.

Figure 5 plots the canonical model of each stage from Table 2 against the train-mean constant.

A.2 Horizon definition

Table 9 compares the trading-day and calendar-day definitions of the horizon on FinCall and MAEC, both at the level of the targets (sessions per window, correlation between the two definitions) and at the level of the Stage-0 results (persistence MSE and HAR-RV R^2_{OOS} under each definition).

A.3 Reproduction study

Table 10 is the first entry of the study, the HTML head over our frozen BGE-M3 chunk sequences on EC under the published split, a gapped split and a ticker-disjoint split with identical labels. Table 11 reruns the five papers with usable code on their own data with the authors' code wherever it runs and a line-by-line port where it

Model	τ=3	τ=7	τ=15	τ=30
Temporal split (n=483)				
Ridge (eGeMAPS)	0.163*	0.156*	0.059*	0.079*
Ridge (WavLM)	0.156*	0.148*	-0.027	-0.008*
Ridge (emotion2vec+)	0.155*	0.129*	0.022*	0.027*
Ridge (WavLM+past-vol)	0.357	0.262	-0.044	0.052*
Ticker-disjoint split (n=484)				
Ridge (eGeMAPS)	0.203*	0.119*	0.137*	0.044*
Ridge (WavLM)	0.178*	0.192*	0.131*	0.026*
Ridge (emotion2vec+)	0.181*	0.185*	0.122*	0.047*
Ridge (WavLM+past-vol)	0.456	0.390	0.267	0.222
Audio-shuffle control of WavLM+past-vol (ticker-disjoint)				
Real audio	0.456	0.390	0.267	0.222
Within-ticker shuffle	0.453	0.373	0.229	0.152
Global shuffle	0.453	0.323	0.164	0.099

Table 8: Audio content (Stage 3). R^2_{OOS} vs. persistence on the Δv target, FinCall-Surprise (test). Audio-only heads stay within 0.10 of the train-mean constant and never beat it by more than 0.02. The head with past volatility is below the past-volatility-only ridge of Table 7 in every cell, and at τ=3 it keeps its score when every test call receives a random other company's audio (bottom block), so past volatility rather than audio content carries it. Asterisks mark Diebold–Mariano significance vs. the Stage-1 ticker-FE GBDT ($p < 0.05$, read the direction from the value). Full head set in Appendix A.

cannot. The released fidelity ledger records every patch and substitution. Table 12 gives the absolute values behind Table 4, and Table 15 the code-availability audit of the seventeen papers.

Same-Company-Same-Signal reproduces. Its TSMixer baseline matches the 76 published cells to a median 10^{-5} from the shipped data alone. Its TMLP text model on the authors' OpenAI embeddings matches per-window means within 0.03 and recovers the paper's finding that embeddings swapped within a company do as well as the real ones. Its training-free baselines on EC and MAEC reproduce all 48 published cells at the notebook's rounding. **HTML** does not. The released driver splits features and labels with three independent unseeded shuffles. With the authors' model classes unchanged on rebuilt inputs, the published recipe (ten epochs, learning rate 2×10^{-5}, raw log-volatility targets) beats the published number only at τ=3 and τ=7 on the shipped code's random split with dropout 0, reaches the published 15- and 30-day values on no split, and on the VolTAGE split every later paper inherited it is worse than persistence at every horizon except τ=3. Its text+audio row reproduces under the shipped code's random split and under no other. **Sawhney et al.** cannot run as released (an unshipped call inventory, price files and lexicons, and a TensorFlow 2.1 stack). Ported, the finance SVR alone gives 0.712/0.370/0.218/0.154, the text and audio branches are untrained after the two and one epochs the scripts prescribe, and the published ensemble is reproduced only when the mixing weights are chosen on the test set, as the released script does, where an untrained branch's near-constant output acts as an intercept on the finance branch. **DialogueGAT**'s corpus

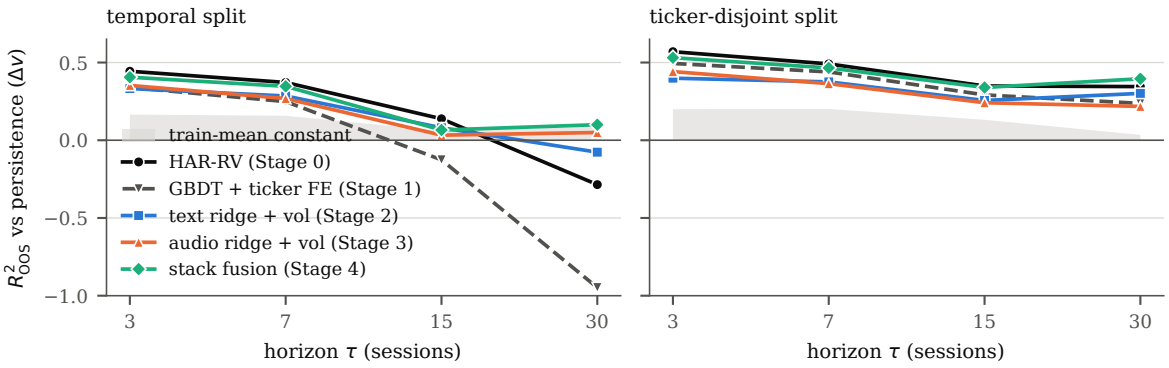


Figure 5: The ladder. R^2_{OOS} vs. persistence on Δv (FinCall test) for the canonical model of each stage. The shaded band is the train-mean constant. No content stage clears HAR-RV on the temporal split, and on the ticker-disjoint split only the fusion stack does, at τ=30. Data from Table 2.

	τ=3	τ=7	τ=15	τ=30
Targets (FinCall, calls ok under both conventions)				
Sessions in calendar window (mean)	2.4	4.9	10.8	21.2
Calls with ≥2 sessions (of 2,433)	1,398	2,433	2,433	2,433
corr($v^{\text{trad}}_{\text{post}}, v^{\text{cal}}_{\text{post}}$)	0.784	0.942	0.963	0.940
corr($\Delta v^{\text{trad}}, \Delta v^{\text{cal}}$)	0.751	0.887	0.904	0.891
Persistence MSE, level-v, temporal test (trading / calendar)				
FinCall	1.270 / 2.071	0.448 / 0.612	0.211 / 0.298	0.115 / 0.152
MAEC	1.774 / 2.553	0.712 / 0.986	0.356 / 0.492	0.201 / 0.263
HAR-RV R^2_{OOS} vs. persistence, level-v, temporal test (trading / calendar)				
FinCall	0.444 / 0.419	0.372 / 0.402	0.138 / 0.275	−0.285 / −0.053
MAEC	0.603 / 0.562	0.486 / 0.502	0.321 / 0.400	0.221 / 0.265

Table 9: Horizon definition. The calendar-day windows of Qin and Yang (2019) hold about 0.7τ sessions, and at τ=3 43% of the calls have at most one post-call session and are excluded. The persistence MSE, the leaderboard's metric, rises by 44–63% at τ=3 under calendar days, so published MSE values are not comparable across definitions. The FinCall temporal τ=30 deficit of HAR-RV against persistence (the COVID-regime cell, §4) holds under both definitions (−0.285 and −0.053). The full model and horizon grid under both definitions is in the supplement.

and labels were never released. Ported onto FinCall's speaker turns under the paper's per-year protocol, it beats persistence at short horizons and collapses on the 2020 split. **KeFVP**'s released script needed five release-defect patches before it would start. Its MAEC rows use sentence embeddings regenerated with the authors' own generator because the released ones were never published, and seven of eight cells fall within one published standard deviation (the eighth, MAEC-16 at τ=15, is the row of their table that does not average to its own stated mean). On EC the released KePt embeddings are unavailable upstream (the linked folder is empty). With BERT-large pooler embeddings from the same generator, which is KeFVP without its knowledge-enhanced pre-training, the mean test MSE is 0.651/0.302/0.194/0.127 against the published 0.610/0.291/0.183/0.114, above all four published means and within one of our repeat standard deviations at only two horizons, so the row is not counted as reproduced.

Under the checks, the 30-session gap removes half to two thirds of KeFVP's MAEC training calls, so those rows cannot separate leakage from data loss and are left to the ledger. DialogueGAT's port (one seed, 42–60 held-out calls per year) shows no consistent direction under either check.

A.4 Audio by bitrate on Earnings25

Table 13 reruns the Stage-3 audio models on Earnings25, pooled and per bitrate stratum, with the random swap as the check.

A.5 Post-cutoff evaluation under the call-date anchor

Table 14 repeats the post-cutoff evaluation with day 0 anchored on the call date on both sides of the cutoff, the anchor every released label set uses. HAR-RV barely moves, but the ridge models that carry a learned past-volatility block collapse before the cutoff under that anchor, because the block absorbs the reaction session that the after-hours rule leaves inside the

Condition (EC)	model	τ=3	τ=7	τ=15	τ=30
paper, published split, published labels	HTML (text, WWM-BERT)	1.175	0.372	0.153	0.133
published split, published labels	HTML head (BGE-M3)	0.769	0.436	0.285	0.223
	persistence	1.485	0.570	0.340	0.202
published split, our labels	HTML head (BGE-M3)	0.760	0.420	0.285	0.214
	persistence	1.826	0.590	0.333	0.183
embargoed temporal, our labels	HTML head (BGE-M3)	0.778	0.448	0.278	0.265
	persistence	1.948	0.629	0.353	0.192
ticker-disjoint, our labels	HTML head (BGE-M3)	**0.773**	**0.503**	**0.394**	**0.291**
	persistence	1.670	0.728	0.418	0.213

Table 10: Reproduced prior model under leakage-proof evaluation (Result Table 6R, first entry). The HTML head reproduces the paper's ordering against price baselines at τ=3 and beats the published number there, but not the published τ=15 and τ=30 values, which lie below the persistence floor of the shipped labels themselves. At τ=30 every condition is worse than persistence. Moving the same head and labels from the published split to a ticker-disjoint split raises test MSE by 2%, 20%, 38% and 36%. The embargoed temporal split costs 2%, 7%, −2% and 24%.

pre-window for before-open calls. On Δv at τ=30 the frozen WavLM+past-volatility model is the only content model whose post-cutoff confidence interval excludes zero under the measured anchor (0.18, CI 0.14–0.24, against HAR's 0.08, and the constant also scores 0.18 there). That is a single-quarter observation without a Diebold–Mariano test and we report it as exploratory.

A.6 Code availability

Table 15 is the code-availability audit of the seventeen papers in this line, regenerated by `ecvol audit code`.

Paper / setting	what ran	ours				published			
		τ=3	7	15	30	τ=3	7	15	30
HTML (Yang et al., 2020), EC, authors' model classes verbatim, inputs rebuilt (WWM-BERT layer −2), driver corrected									
code: random 70/10/20, dropout 0	text	0.660	0.336	0.253	0.160	1.175	0.372	0.153	0.133
paper: chronological 7:1:2, dropout 0.5	text	0.972	0.439	0.406	0.248				
published split (VolTAGE), dropout 0.5	text	1.231	1.122	1.063	1.479				
persistence on the shipped labels	n/a	1.485	0.570	0.340	0.202				
code: random 70/10/20, dropout 0	text + 27 Praat	0.658	0.347	0.244	0.165	0.845	0.349	0.251	0.158
paper: chronological 7:1:2, dropout 0.5	text + 27 Praat	0.876	0.520	0.291	0.239				
published split (VolTAGE), dropout 0.5	text + 27 Praat	2.701	1.195	1.436	1.411				
Same-Company-Same-Signal (Yu et al., 2025), DEC, authors' `run.py`, per-window means over 19 rolling quarters									
TSMixer (price history only)	76 cells	0.561	0.252	0.210	0.257	0.561	0.252	0.211	0.256
TMLP on OpenAI embeddings	76 cells	0.608	0.285	0.215	0.202	0.587	0.283	0.211	0.205
TMLP, embeddings shuffled within ticker	76 cells	0.572	0.276	0.213	0.209	0.565	0.273	0.204	0.200
TMLP, embeddings shuffled across all	76 cells	0.646	0.315	0.259	0.249	0.637	0.319	0.255	0.247
Same-Company-Same-Signal, EC / MAEC, authors' notebook, training-free, augmented same-ticker earnings history									
EC, Aug_STPEV	no training	0.569	0.293	0.201	0.122	0.569	0.293	0.201	0.122
MAEC-15, Aug_STPEV	no training	0.443	0.214	0.130	0.112	0.443	0.214	0.130	0.112
MAEC-16, Aug_STPEV	no training	0.470	0.238	0.168	0.113	0.470	0.238	0.168	0.113
KeFVP (Niu et al., 2023), authors' `final_series_infer.py` (five release-defect patches), 10 repeats									
MAEC-15, regenerated BERT-base embeddings	text + price	0.419	0.185	0.123	0.086	0.418	0.187	0.122	0.087
MAEC-16, regenerated BERT-base embeddings	text + price	0.442	0.288	0.363	0.188	0.445	0.279	0.303	0.177
EC, regenerated BERT-large embeddings (no KePt)	text + price	0.651	0.302	0.194	0.127	0.610	0.291	0.183	0.114
Sawhney et al. (2020b), EC, PyTorch port of the released scripts (their labels, split, branches); 5 seeds									
finance SVR (30 past volatilities)	port	0.712	0.370	0.218	0.154				
text BiLSTM, their 2 epochs	port	2.135	1.580	1.384	1.401				
aligned audio, their 1 epoch	port	6.514	5.295	5.368	5.130				
ensemble, (α, β) tuned on test (theirs)	port	0.639	0.298	0.175	0.102	0.601	0.308	0.181	0.119
ensemble, (α, β) tuned on validation	port	0.686	0.324	0.187	0.117				
persistence on their labels	n/a	1.491	0.563	0.345	0.205				
DialogueGAT (Sang and Bao, 2022), PyG port on FinCall turns (role-typed speakers); published cells are on their unreleased corpus									
FinCall 2019 (their 2015 protocol)	port	0.694	0.320	0.206	0.134	0.453	0.324	0.190	n/a
persistence	n/a	1.429	0.566	0.289	0.139				
FinCall 2020 (their 2016 protocol)	port	0.621	0.350	0.392	0.270	0.455	0.288	0.181	n/a
persistence	n/a	1.409	0.335	0.184	0.101				
FinCall 2021 (their 2017–18 protocol)	port	0.794	0.344	0.182	0.082	0.409	0.289	0.204	n/a
persistence	n/a	1.301	0.465	0.225	0.118				

Table 11: Reproductions with the authors' own code and data (Result Table 6R-F, T6R.3). Every cell regenerates from one `ecvol reproduce` command, and the fidelity ledger lists what is verbatim, patched, substituted or reason-coded per paper. Three of the five reproduce. TSMixer matches to 10^{-5}, TMLP within single-seed noise (including the paper's finding that real and within-company shuffled embeddings score alike), and KeFVP on MAEC within one published standard deviation on seven of eight cells with regenerated sentence embeddings. On EC, whose released KePt embeddings are unavailable upstream, the regenerated BERT-large row is above all four published means and within one of our repeat standard deviations at only two horizons, and is not counted. HTML's text-only recipe does not reach its published long-horizon cells on any split, while its text+audio row reproduces at $\tau \geq 7$ under the shipped-code condition (random split, dropout 0) and on no other. Sawhney's published ensemble is reachable only when the mixing weights are tuned on the test set, as the released code does, with untrained text and audio branches acting as an intercept on the finance SVR.

Model / condition	n_{tr}	n_{te}	τ=3	7	15	30
HTML, authors' classes (EC; seeds 0–2)						
text, shipped-code random split (anchor)	≈390	112	0.660	0.336	0.253	0.160
text, embargoed	285	101	0.726	0.347	0.246	0.178
text, ticker-disjoint	386	109	0.777	0.522	0.393	0.291
persistence on the ticker-disjoint test set	n/a	109	1.452	0.550	0.327	0.188
text+audio, anchor	≈390	112	0.658	0.347	0.244	0.165
text+audio, embargoed	285	101	0.684	0.345	0.232	0.186
text+audio, ticker-disjoint	386	109	0.789	0.500	0.394	0.281
transcript shuffle, anchor split, the 97 test calls whose company has another call						
text, real transcript	≈390	97	0.683	0.348	0.262	0.165
text, another call of the *same* company	≈390	97	0.693	0.361	0.269	0.169
text, a call of a *different* company	≈390	97	0.745	0.386	0.318	0.203
text+audio, real / same company / different company, τ=3	≈390	97	0.675 / 0.692 / 0.784			
Sawhney et al., port, ensemble with (α, β) tuned on validation (EC; 5 seeds)						
their split (anchor)	328	110	0.686	0.324	0.187	0.117
embargoed	283	100	0.719	0.306	0.184	0.107
ticker-disjoint	383	109	0.775	0.416	0.263	0.161
its own finance SVR, ticker-disjoint	383	109	0.700	0.365	0.256	0.158
Same-Company-Same-Signal, authors' `run.py` (DEC; seed 2021; means over 19 rolling quarters)						
TMLP (text), paired anchor	900	30	0.591	0.288	0.224	0.211
TMLP (text), ticker-disjoint	600	30	0.660	0.354	0.280	0.267
TMLP (text), embargoed (vs anchor 0.608 / 0.285 / 0.215 / 0.202)	895	90	0.585	0.287	0.216	0.208
TSMixer (price), paired anchor	900	30	0.519	0.241	0.206	0.252
TSMixer (price), ticker-disjoint	600	30	0.556	0.254	0.194	0.233
persistence on the held-out calls	n/a	30	1.258	0.457	0.252	0.224
KeFVP, authors' script (MAEC, regenerated BERT-base embeddings; 3 repeats)						
MAEC-15, paired anchor	535	48	0.470	0.192	0.145	0.114
MAEC-15, ticker-disjoint	353	48	0.458	0.199	0.150	0.113
MAEC-16, paired anchor	980	92	0.416	0.248	0.347	0.186
MAEC-16, ticker-disjoint	661	92	0.378	0.241	0.312	0.178
persistence on the MAEC-16 held-out calls	n/a	92	0.622	0.265	0.170	0.116

Table 12: The reproduced models under our controls (Result Table 6R-C, T6R.4). Each block keeps the configuration of Table 11 and changes only the split. Paired anchors score the unchanged model on exactly the ticker-disjoint test calls. Removing same-company history costs the three text models 12–82% (HTML, Sawhney, TMLP, where TMLP is worse in 16–18 of 19 quarters) and leaves both price-dominated models unchanged (TSMixer, KeFVP), although the ticker-disjoint training sets are as large (EC) or cut by the same third (DEC, MAEC) in both groups. The 30-session embargo moves HTML, Sawhney and SCSS by at most 13% in either direction. KeFVP's embargoed MAEC re-splits keep only a half or a third of its training calls, cannot separate leakage from data loss, and are reported in the ledger rather than here. DialogueGAT (one seed, 42–60 held-out calls per year) shows no consistent direction and is also left to the ledger.

Stratum	calls (test)		τ=3	τ=7	τ=15	τ=30
All	514 (95)	real	0.435	0.328	0.171	0.220
		global shuffle	0.454	0.327	0.141	0.206
		increment	**−0.019**	**0.001**	**0.030**	**0.014**
64 kbps	281 (39)	real	0.370	0.250	0.109	0.310
		global shuffle	0.383	0.266	0.133	0.303
		increment	**−0.013**	**−0.016**	**−0.024**	**0.007**
≤24 kbps	232 (55)	real	0.343	0.328	0.174	0.145
		global shuffle	0.443	0.354	0.174	0.128
		increment	**−0.100**	**−0.026**	**−0.001**	**0.018**

Table 13: Audio on Earnings25 by bitrate stratum (T9.4). R^2_{OOS} vs. persistence on Δv for the ridge on WavLM plus past volatility, ticker-disjoint test. With one call per company the same-company swap is the identity, so the random swap is the check. Call counts are inventory calls with audio at that bitrate (one call at 32 kbps sits in neither stratum). The heads see the 475 admitted, priced calls, and the test count is their ticker-disjoint segment. The audio increment never exceeds 0.03, is negative in 7 of the 12 cells, and has no consistent sign across strata. The 64 kbps stratum is not better than the ≤24 kbps one. The full grid (eGeMAPS, emotion2vec+, fusion, MLP heads, all targets) is in `result_table_3_earnings25.csv`.

Head	seg.	τ=3	τ=7	τ=15	τ=30
HAR-RV	in	$0.44 \to 0.46$	$0.37 \to 0.41$	$0.14 \to 0.21$	$-0.29 \to -0.28$
	post	$0.60 \to 0.56$	$0.54 \to 0.51$	$0.35 \to 0.33$	$0.08 \to -0.12$
text + past-vol	in	$0.33 \to -1.71$	$0.28 \to -3.12$	$0.08 \to -3.62$	$-0.08 \to -0.47$
	post	$0.31 \to 0.28$	$0.33 \to 0.27$	$0.23 \to 0.15$	$0.12 \to -0.09$
WavLM + past-vol	in	$0.36 \to 0.26$	$0.26 \to -1.02$	$-0.04 \to -2.01$	$0.05 \to -0.56$
	post	$0.50 \to 0.43$	$0.43 \to 0.32$	$0.15 \to 0.06$	$0.18 \to 0.05$
fusion + past-vol	in	$0.29 \to -1.73$	$0.22 \to -3.13$	$0.04 \to -0.34$	$-0.11 \to -0.45$
	post	$-0.83 \to -9.97$	$0.32 \to -2.29$	$0.19 \to 0.02$	$0.01 \to -0.28$

Table 14: Anchor sensitivity. The same frozen heads with day 0 on the call date, the anchor of the released label sets (Δv, R^2_{OOS} vs. persistence). Each cell reads measured $\to$ call-date. The ordering on level-v is in the CSV.

Paper	venue	probe (date)	verdict	reason
Qin & Yang, What You Say and How You Say It Matters	ACL 2019	200, 0 code files, pushed 2022-08-04 (2026-10-09)	partial	dataset released (Drive, 5-part zip); no model code in the repository
Yang et al., HTML	WWW 2020	200, 15 code files, pushed 2024-01-05 (2026-10-09)	partial	model classes run verbatim; the released driver imports TensorFlow-1 set_random_seed and torchtext and splits features and labels with independent unseeded shuffles; inputs (WWM-BERT .npy, Praat features) behind dead Drive links, rebuilt; run with a corrected driver (T6R.3)
Sawhney et al., multimodal financial forecasting	ACM MM 2020	200, 112 code files, pushed 2020-10-26 (2026-10-09)	partial	TF 2.1 / Keras 2.3.1 / tensorflow-addons 0.8.3 pins; unshipped call inventory, price files and two lexicons; text-feature script returns after one record; no multi-task loss in the released code; ported line by line to PyTorch (T6R.3)
Sawhney et al., VolTAGE	EMNLP 2020	200, 5 code files, pushed 2021-11-16 (2026-10-09)	partial	ships the EC split/label files (byte-identical to KeFVP's) and GCN code; embeddings as .pkl; not ported
Niu et al., KeFVP	EMNLP Findings 2023	200, 42 code files, pushed 2024-01-28 (2026-10-09)	partial	requirements.txt does not install as written; the script starts only after five release-defect patches (missing set_seed, package init importing undefined names, a dropped class header, an unreleased 'latent' package, plotting imports); MAEC embeddings never released (regenerated with the authors' generator); EC KePt pickle Drive-only (T6R.3)
DialogueGAT	Findings of EMNLP 2022	200, 6 code files, pushed 2023-06-12 (2026-10-09)	partial	MIT code, tau<=15 only; training pickle, SeekingAlpha corpus and CRSP labels never released; DGL has no wheels for torch 2.11; ported (PyG) onto FinCall turns (T6R.3)
Yu, Liu & He, Same Company, Same Signal	Findings of ACL 2025	200, 38 code files, pushed 2026-08-13 (2026-10-09)	runnable	MIT; ships DEC with features and rolling masks; PEV/STPEV to 3 decimals, TSMixer 76 cells to 1e-5, TMLP 228 cells within single-seed noise on the authors' embeddings
Yang et al., FinTrust	ACL 2023 short	200, 4 code files, pushed 2024-01-22 (2026-10-09)	partial	perturbation harness + 3/7/15/30 data; complementary control axis, not a volatility model
Sound of Risk	2025	200, 0 code files, pushed 2025-08-26 (2026-10-09)	none	repository holds figures only, zero source files; proprietary 1,795-call corpus
DeFVP	2024	409 (2026-10-09)	none	repository exists but is empty (size 0 KB, never pushed; API returns 409)
ECHO-GL	2024	200, 3 code files, pushed 2024-02-26 (2026-10-09)	none	README states the code 'can not run at present'; target is direction, not volatility
ECC Analyzer	2024	no_url	none	no code released; GPT-4 Turbo + OpenAI embeddings
RiskLabs	2024	no_url	none	no code released; GPT-4/3.5
NumHTML	AAAI 2022	no_url	none	paper promises code; no URL given
GNA-Vol		no_url	none	artefact URL returns 404 (checked 2026-08-23)
AMA-LSTM		no_url	none	10-byte stub repository, unattributed
AT-FinGPT		no_url	none	paywalled; no artefact

Table 15: Code-availability audit of the ECC volatility literature (T6R.2/T6R.3, regenerated by `ecvol audit code` with a live GitHub probe on the date shown). One of seventeen papers ships code that runs as released. Seven ship code with a broken environment, missing data, or defects that must be repaired before it starts (HTML's driver imports TensorFlow-1 symbols and decorrelates features from labels, and KeFVP needs five release-defect patches). Nine ship nothing usable. The reason column records what the faithful-reproduction sprint (Table 11) did with each.